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# National Social Life, Health, and Aging Project (NSHAP): Round 1, [United States], 2005-2006

Data Release Manual

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**National Social Life, Health, and Aging Project (NSHAP): Round 1, [United States], 2005-2006**

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# National Social Life, Health and Aging Project (NSHAP)

***Release 3.0***

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# CONTENTS

# CHAPTER ONE

# NSHAP PUBLIC DATA RELEASE

**Important:** This release includes the first batch of data from Wave 3. Some items are still undergoing quality control checks and data cleaning/coding, and are not included—these will be added to a subsequent release. If you have any questions about the Wave 3 data (including items not yet included), please email them to the nshap-data discussion list at nshap-data@lists.uchicago.edu (see *Reporting A Problem* for further details).

## 1.1 Introduction

This document is intended to accompany the public release of the National Social Life, Health and Aging Project (NSHAP) dataset, which is being made available through the National Archive of Computerized Data on Aging (NACDA). It describes the contents of the distribution, and provides information necessary to analyze the data.

NSHAP is a population-based study of health, social life, and well-being among older Americans. A nationally-representative probability sample of community-dwelling individuals ages 57-85 was selected from households across the U.S. screened in 2004. African-Americans, Latinos, men and the oldest-old (75-84 years at the time of screening) were over-sampled. In-home interviews were conducted in English and Spanish by professional interviewers from the National Opinion Research Center (NORC) between July 2005 and March 2006, yielding a total of 3,005 respondents (1,455 men and 1,550 women). The weighted sample response rate was 75.5%.

Data collection for Wave 1 included a face-to-face interview (including a brief self-administered questionnaire), in-home collection of a broad panel of biomeasures, and a leave-behind questionnaire (the response rate for the leave-behind was 84%). Topics covered in the interview and leave-behind included self-reported health, physical function and morbidity, social networks, social support, marital history and intimate partnerships, sexuality, medication use, and demographic information. The panel of biomeasures included: weight, waist circumference, height, blood pressure, smell, saliva collection, taste, a self-administered vaginal swab for female respondents, "Get Up and Go", distance vision, touch, Orasure (a test for HIV), and blood spots. Following completion of the interview, the interviewer answered a series of questions concerning the respondent's physical appearance, physical and sensory function, home and neighborhood, and the overall context of the interview.

A second wave was conducted from August 2010 through May 2011, during which Wave 1 respondents were reinterviewed. An attempt was also made to interview individuals who were sampled in Wave 1 but declined to participate (referred to as *Wave 1 non-respondents*). In addition, spouses or coresident partners were also interviewed using the same instruments as the main respondent. The resulting Wave 2 dataset contains 3,377 total respondents, including 957 couples.

Wave 3 was conducted from September 2015 through November 2016. Two-thousand four-hundred and nine surviving Wave 2 respondents were reinterviewed, and a *New Cohort* consisting of adults born between 1948 and 1965 together with their spouses or coresident partners was added (n = 2,368 total). As with the *Original Cohort*, the New Cohort was obtained from a probability sample of U.S. adults meeting the age criteria, with African-Americans and Hispanics sampled at a higher rate. All together, 4,777 respondents were interviewed in Wave 3.

The COVID-19 substudy is a brief self-report questionnaire that probes how the coronavirus pandemic has changed older adults' lives. It was administered during Fall 2020 via web survey, phone interview, or paper-and-pencil mail-back instrument. The questionnaire was designed for respondents in the NSHAP main study, on whom considerable background information is already available. All respondents to the NSHAP Round 3 (2015—16) main study were invited to respond to the COVID-19 substudy questionnaire, and thus it was limited to assessing specific domains in which respondents may have been affected by the coronavirus pandemic, and includes: (1) COVID experiences, (2) health and health care, (3) job and finances, (4) social support, (5) marital status and relationship quality, (6) social activity and engagement, (7) living arrangements, (8) household composition and size, (9) mental health, (10) elder mistreatment, (11) health behaviors, and (12) positive impacts of the coronavirus pandemic. Questions about engagement in racial justice issues since the death of George Floyd in police custody were also added to facilitate analysis of the independent and compounding effects of both the COVID-19 pandemic and reckoning with longstanding racial injustice in America.

## 1.2 Modular Design of Interview

### 1.2.1 Wave 1

In order to keep the average time spent in the home from exceeding an acceptable limit, the following steps were taken:

1) some questions were relegated entirely to the leave-behind
2) some biomeasures were collected only among a randomized subset of respondents, and
3) some questions were asked of a randomized subset of respondents during the interview, with the others being asked in the leave-behind

In order to implement (2) and (3), the affected items were organized into the following 6 modules:

A) questions on social support, elder mistreatment and physical contact
B) questions on children/grandchildren, sexual interest and physical health
C) blood spots
D) Orasure
E) biomeasures of physical and sensory function

Respondents were then randomly assigned to one of 6 interview paths, with the path determining whether they received modules A and B as part of the main interview or in the leave-behind, and whether they received modules C-E during the interview or not at all. Please see the included questionnaire for an in-depth description of the individual paths.

In cases where a particular question appeared in the interview for some and in the leave-behind for others, we have combined the resulting data into a single variable to facilitate analysis. Nonetheless, this difference in the mode of administration may have consequences for the distribution of the responses requiring that they be treated differently for the two groups (at the very least, users will want to compare the distribution across the two groups before including the variable in further analyses). This may be done easily by using the variable `path` which indicates which path the respondent was assigned to. For example, to compare the distributions between interview and leave-behind for the variable `famrely` (corresponding to the question "How often can you rely on [your family] for help…") which was asked in the leave-behind for paths 5 and 6, one could use the following Stata commands:

```
. recode path 1/4=0 5/6=1, gen(paths56)
. tab famrely paths56, col
```

In addition, for convenience any time a question was asked in the leave-behind for either some or all of the respondents a note indicating this has been attached to the corresponding variable. For example, here is the note attached to `famrely`:

```
. notes famrely

famrely:
  1.  asked in leave-behind for path(s) 5 and 6
```

### 1.2.2 Wave 2

In Wave 2, the instrument was simplified to facilitate both administration and analysis. In particular, all questions were asked entirely in the in-person interview or the leave-behind (no mixed-mode questions). In addition, partner history items regarding relationship details and sexual experience were combined into the same modules. As in Wave 1, selected biomeasures were administered to a randomized subset of respondents in cases where this provided adequate power for likely analyses. The variable `path` may be used to determine the various subsamples.

## 1.3 Sample Design

### 1.3.1 Wave 1

NSHAP used a complex, multi-stage area probability sample. The sample design was integrated with that of the Health and Retirement Study (HRS) in order to share the costs of frame development and sample screening between the two projects. A detailed description of the sample design is available [OMUIRCHEARTAIGH2009].

#### Survey Weights

Two sets of weights are available. Respondent-level weights representing the inverse probability of selection are contained in the variable `weight_sel`. A second set of weights incorporating a non-response adjustment based on age and urbanicity is contained in the variable `weight_adj`. Both sets of weights are scaled to sum to the final sample size (3,005).

#### Design-Based Variance Estimates

Researchers wishing to compute design-based variance estimates may use the variables `stratum` and `cluster`. These variables were constructed from the original sampling units for the purpose of variance estimation; the former may be treated as (pseudo) strata and the latter as (pseudo) Primary Sampling Units (PSUs). These may be used with Stata's commands for survey data analysis by issuing the following declaration:

```
. svyset cluster [pweight=weight_adj], strata(stratum)
```

The core file has already been setup using the command above, which you may verify for yourself by typing:

```
. svyset
```

### 1.3.2 Wave 2

In Wave 2, some partners of Wave 1 respondents would not have been age-eligible for Wave 1 (i.e., they were not born between 1920 and 1947). To restrict inferences to the population of Wave 1 age-eligibles, analysts may use the variable `ageelig` in the Wave 2 core file. For example, in Stata you would use the following `svy` prefix command:

```
. svy, subpop(ageelig): [command]
```

where [command] is replaced with an appropriate analysis command.

Note that the weights (`weight_sel` and `weight_adj`) and sample design variables (`stratum` and `cluster`) in the Wave 2 core file may be used as described above for Wave 1.

A detailed description of the Wave 2 sample augmentation and procedures for estimation is available [OMUIRCHEARTAIGH2014].

### 1.3.3 Wave 3

Although the New Cohort added in Wave 3 was selected from a national frame, it was not the same as that used to select the Original Cohort (the Original Cohort was sampled from the Health and Retirement Study (HRS) sampling frame, while the New Cohort was sampled from NORC's national frame). As with the Original Cohort, African Americans and Hispanics were oversampled in the New Cohort to provide adequate cases for subgroup analyses. Spouses or coresident partners of sampled respondents were also included.

At the moment, non-response adjusted weights and other sample design variables (strata and Primary Sampling Unit) are not available. These will be added in a subsequent release.

## 1.4 Variable Coding

A consistent coding scheme has been used for all variables. In assigning numeric codes, the following rules were used:

1) no/yes responses are always coded 0/1

2) Whenever an item has a natural directionality, this is reflected in the coding even if the responses appeared differently in the actual questionnaire. For example, although the responses to the questions on self-rated health were presented in the order “Excellent”, “Very Good”, etc., the variables in the dataset are coded as:

```
1 poor
2 fair
3 good
4 very good
5 excellent
```

This has been done to facilitate analysis. In all cases, the definitive source for the item as it was presented to the respondent is the questionnaire.

3) Whenever a response set contains a natural zero, this is also reflected in the coding. For example, the variable `hlthvis` (corresponding to the question “During the past 12 months, how many times have you seen a doctor or other health care professional. . . ”) is encoded like this:

```
0 none
1 one
2 2-3
3 4-9
4 10-12 (about once a month)
```

(continues on next page)

(continued from previous page)

```
5 13-20
6 20-30 (about twice a month)
7 30 or more
```

This has also been done to facilitate analysis. In cases where a natural zero is not present, integers beginning with the number one are used.

4) No system missing values (i.e., sysmiss or .) appear in the data files. Instead, specific reasons for missingness have been identified and labeled using Stata's extended missing values (i.e., .a, .b, etc.). The following missing value codes are used consistently throughout all data files:

```
.a refused
.b don't know
.c not applicable
.d no answer
.e not returned
.f missing in error
.g insufficient data
.h incomplete interview
```

Because some data analysis applications do not support extended missing values (e.g., R, SPSS), a companion version of each data file is provided in which the extended missing values have been recoded to negative integers (i.e., .a -> -1, .b -> -2, .c -> -3, etc.). These files have names ending with "noextended," and are otherwise equivalent to their counterparts.

All cases where an item was skipped as specified by the questionnaire are marked as "not applicable". The categories "no answer" and "not returned" apply only to responses obtained from the leave-behind. One respondent broke off the interview and did not complete it; all variables from that point forward for that individual are coded "incomplete interview". Finally, "missing in error" is used whenever a value should be available but is not (e.g., due to interviewer error, technical error, or some other type of problem); in these cases a note usually appears in the file explaining what happened.

Very few derived variables have been included in this release. However, when a derived variable cannot be computed because of missing information among the constituent items, this is usually coded as "not applicable". Note also that whenever a derived variable is provided, the Stata code used to derive that variable is also provided in a note attached to the variable.

## 1.5 Date Variables

The NSHAP interview included several questions about the dates on which specific events occurred, such as the dates on which a marriage began or ended (in the marital/cohabiting history section) or the dates of first or last sex with a specific partner (in the sexual partner history section). In all cases, only the month and year (not the actual date) were collected. Since in some cases respondents provided only partial information (e.g., just the year), each date item is stored in a pair of numeric variables—one for the month and one for the year.

In order to permit users to work with historical dates relative to the date of the interview, we have also included the month and year in which the interview was conducted (the actual date of the interview has been withheld to protect respondent confidentiality). Fifteen respondents began and finished the interview in different months; in these cases, we have defined the interview date to be the date on which Section 3 (marriage, cohabitation, and sexuality) was started. Since the first date variables appear in Section 3, calculations using this date most closely reproduce those made during the interview by the CAPI application. Only 8 respondents started Section 3 in a different month from the month in which they began the interview, and in four of these cases the difference between the start of the interview and the start of Section 3 was only one month.

Since the date of the interview is nonmissing in all cases, it is stored in a single numeric variable as the number of months since January 1960.

## 1.6 Long-format Files

Data consisting of multiple values per respondent are provided in *long format* to save space and to facilitate analysis.

### 1.6.1 Social Networks

The network file contains one record for each person identified on the network roster. A within-respondent unique ID is provided by the variable `lineno`; this is identical to the original line number from the actual roster. Respondents who refused to participate in the roster or who did not identify anyone are not represented in this file.

An effort has been made to purge this file of any duplicates (i.e., persons listed twice on the roster) or persons discovered elsewhere in the interview to be deceased.

Respondents with a current spouse, cohabiting partner, or other "romantic, intimate, or sexual partner" were asked to identify this person on the roster if they had not already done so. Thus, such a person should always be represented in the network file, and may be identified using the variable `spcurrent`. Later on, respondents completed up to two loops through the section on sexual behaviors and problems (Section 3C), one of which was always with respect to the spouse, cohab, or partner identified on the roster. In most cases this corresponds to the first loop through Section 3C, but in a few cases it corresponds to the second. This is indicated by the variable `sploop` in the core data file (immediately preceding the variables containing the data from Section 3C). Users may use this variable to match the spouse, cohab, or partner identified in the roster with the corresponding information on sexual behavior and problems.

### 1.6.2 Wave 1 Marital/Cohabiting History

The marital history file contains one record for each marriage or cohabitation identified in Section 3A. A within-respondent unique ID is provided by the variable `scid`; these begin with 100 for the marriages and 201 for the cohabs. The order of these IDs matches the order in which the relationship was identified when completing the section. Respondents who did not identify any spouses or cohabs are not represented in this file.

In several cases, a current cohab or spouse (or ex-spouse) identified in the marital history was then identified again in the cohab history. We have attempted to identify all such cases, and have removed the corresponding duplicate records from the file. Note that to avoid confusion, we did not renumber the scids; as a result, there may be gaps in the sequence of scids within respondents.

Although we have removed all duplicates that we can confirm based on auxiliary information (e.g., names, interview notes, etc.), there remain some instances in which the dates within the marital and cohabiting histories are inconsistent or suggest the possibility of additional duplicates (see the notes in the marital history file for further information). Because we have no further information to help explain these cases, we have left the data as recorded so that the analyst may make his or her own decisions about how to handle them. When doing this, one should bear in mind the possibility that the respondent(s) may have reported one or more dates incorrectly.

Finally, note that we did not explicitly collect information on whether a respondent married or cohabited with the same individual more than once; only if the relationship occurred or persisted within the past 5 years did we ask the respondent to identify the partner. Thus, it is possible that two distinct marriages or cohabitations (represented by two different records in the marital history file) involved the same spouse or partner. It is also possible that an instance in which a respondent cohabited with the same person off and on over a long period of time was recorded using only the initial start and final end date (i.e., using only a single record in the marital history file); this possibility should be considered when deciding how to handle apparent duplicates or inconsistencies in individual histories.

> **Warning:** Because some respondents refused and/or answered "don't know" to certain items within the marital/cohab history, histories constructed using the data in this file may in some cases be incomplete (see the notes at the top of the marital history codebook for more information).

### 1.6.3 Wave 1 Sexual Partners

Information on the timing of specific sexual partners from Section 3A (Boxes A and B) is provided in a long format file containing one record for each sexual partner identified. A within-respondent unique ID is provided by the variable `spid`; this ID starts with 301 for partners previously identified during the marital/cohabiting history (i.e., these partners are also either spouses or cohabs), and 401 for additional sex partners within the past 5 years (see vars `aotrsex` and `sexlt5yr`). The order of these IDs matches the order in which the relationship was identified and/or described when completing this section. Respondents who have not been married or cohabited within the past 5 years and have no sexual partners within the past 5 years are not represented in this file.

For convenience, we have included in the core file a variable named `sexlastyr` indicating whether the respondent had sex with at least one partner during the 12 months preceding the interview. This variable was computed based on the sex partner history, and may be reproduced using the sex partners file in conjunction with the variables `sec3_start`, `aotrsex`, and `sexlt5yr` in the core file.

Information mapping the partners described in Boxes A and B to the individuals identified in the network roster and/or in the marital/cohabiting history is not included in the current dataset. This information was spread across multiple locations in the raw file, and is currently being combining and verified. It will be included in a subsequent release.

> **Warning:** Because some respondents provided an incomplete marital/cohabiting history and/or did not answer `aotrsex` or `sexlt5yr`, summaries constructed using the data in this file may in some cases be incomplete.

### 1.6.4 Wave 2 Partner History

The partner history file contains one record for each marriage, cohabitation, or romantic relationship identified in Section 6A, including a current partner in Wave 2 but excluding the partner from Wave 1. Wave 2 simplified the questionnaire by combining sexual histories into the marital/cohabitation module. Thus, the sexual history of spouses and cohabitations are naturally associated with the other partner history information. Many of these variables are derived from answers to Section 6A and thus do not have a direct mapping to the Wave 2 questionnaire.

We have retained the variable names and coding conventions from the Wave 1 Marital/Cohabiting History and Sexual Partners files for convenience.

> **Warning:** Because some respondents refused and/or answered "don't know" to certain items within the marital/cohab history, histories constructed using the data in this file may in some cases be incomplete (see the notes at the top of the marital history codebook for more information).

A within-respondent unique ID is provided by the variable `scpid_w2`; these begin with 2000 and their sequence matches the order in which the relationship was identified when completing the section. Respondents who did not identify any spouses or cohabs are not represented in this file.

Information mapping partners to the individuals identified in the network roster and/or in the partner history is included in the variable `lineno` that identifies the roster line number. If partners are not network members, this variable is set to "not applicable".

**Warning:** We have identified 13 respondents who gave identical line numbers for two partners. In many cases we suspect these duplicates refer to the same individual. A subsequent update will resolve this issue.

### 1.6.5 Wave 2 Disposition of Wave 1 Partner

We created another file to provide information derived from Section 6A items regarding the partner from Wave 1. The variable `mstatus_w1` contains information from Wave 1 used to direct the respondent to appropriate questions. The variable `mstatus` contains the final disposition of the Wave 1 spouse.

Like the Wave 2 Partner History file, these data use variable and coding conventions similar to the Wave 1 Marital/Cohabiting History file.

One use of this file is to update the information about the Wave 1 partner in the Wave 1 Marital/Cohabiting History and Sexual Partners files before combining these data with the Wave 2 Partner History file to create a complete history.

### 1.6.6 Medications

The medications file contains one record for each item listed in the medications log. A within-respondent unique ID is provided by the variable `drug_order`; these begin arbitrarily at 100 to distinuish them from the original "line number" on which the medication was recorded (their order *does* match the order in which the medications were recorded). Respondents who did not report taking any medications or who refused to participate in this module are not represented in this file (see variable `drugs_intro` in core file).

Note that for the protection of our respondents, we have excluded those medications which are used primarily to treat persons with HIV. This affects only 6 respondents (13 records total).

When possible, each record has been matched to a specific drug in the Multum database; in such cases, the corresponding Multum ID and Multum drug name are provided. Because Multum's coverage of nutritional products and alternative medicines is not complete, and because some respondents reported foreign drugs which are not included in the Multum database, several specific items that were reported could not be matched to a Multum drug name. To handle these, an additional set of drug names was developed as an extension (i.e., non-overlapping) to the Multum database. All specific items that were not matched to a Multum drug name were matched to one of these additional names, as recorded in the variable `multum_extended_drug_name`.

Of the 15,389 medications recorded, 15,224 have been identified and matched to either a Multum drug name or a *Multum-extended drug name*. In 78 of the remaining cases, instead of recording a specific drug name the interviewer recorded only a therapeutic objective (e.g., "blood pressure medication"); these have been coded in the variable `drug_coded`. The remaining 87 entries were unrecognizable or otherwise ambiguous; these have been marked "uncodeable".

Multum classifies each drug into one or more therapeutic categories, and we have placed our extended drug names into the same categories (one new category called "other supplements" was added to accommodate some of the extended names). These categories are organized into a three level hierarchy, as described in the included file named `multum_hierarchy.txt`. Thus, for each specific category into which a drug is classified, there may be one or two levels above that category in the hierarchy. This information is contained in the variables `multum_cat#_l#`, where `cat#` refers to each of the lowest level categories into which the drug is classified and `l#` refers to the level of that category (and of each of the categories above it) in the database. In the case of combination drugs, information on the therapeutic classification of the individual components is also provided in the variables `indirect_multum_cat#_l#`.

For convenience, we have included in the core file separate counts of the total number of drugs reported, the total number of nutritional products or alternative medicines, and the total number of unidentified drugs (the vars `np_count` and `unident_count` may be subtracted from `drugs_count` to obtain the total number of identified medications *excluding* nutritional products and alternative medicines). We have also constructed a variable for each of the observed therapeutic categories indicating whether the respondent reported taking one or more medications in that category. These variables are derived from the information in the medications file, and so are guaranteed to be consistent with it.

## 1.7 Direct Download Dataset

Starting with version 3.0, data files are now available for immediate download to registered NACDA users. These files are identical to those in the full data release with the following exceptions:

1) 181 Wave 2 respondents who are not part of the original NSHAP cohort (born between 1920 and 1947) have been excluded. These spouses or cohabiting partners of age-eligible NSHAP respondents are typically excluded from analyses (except analyses focusing explicitly on couples).
2) Information permitting the identification of couples (i.e., two respondents within the same household) has been excluded. Note that since both members of a couple are always included in the same pseudo-PSU, obtaining *Design-Based Variance Estimates* is still possible.
3) Some responses for the variables `degree_coded` (highest educational degree), `race`, `hearn` (household income) and `hsassets` (household assets) have been collapsed into larger categories.
4) Variables containing information on the following topics have been excluded:
    - Elder mistreatment
    - Sexually-transmitted infections
    - Marital infidelity
    - Detailed information about a second sexual partner during the last 5 years (this information was only collected in Wave 1); complete information about the most recent sexual partner *is* included.
5) Detailed information about lifetime marital history and sexual partner history has been excluded.
6) Information about individual medications has been excluded; indicator variables for each Multum medication class are included.

These modifications have been carefully chosen to simplify the dataset and to protect respondent confidentiality while still permitting most of the analyses for which the data are used. Those who wish to utilize the excluded items may do so by requesting the full dataset from NACDA.

## 1.8 Protection of Confidentiality

In order to preserve the confidentiality of our respondents, the following items have been excluded from the public release:

- all direct identifiers (including date of birth and names from network roster and marital/partner history)
- actual date and location of interview (month and year are included), language in which interview was conducted, and description of exactly who was present during interview (presence of 3rd party is included)
- all references (either direct or indirect) to geographical location
- all open-ended responses (e.g., further specification of "Other")
- interviewer's physical description of the respondent
- HPV results
- HIV status (including both test results and self-reported items) and all medications primarily used to treat HIV

## 1.9 Reporting A Problem

> **Warning:** It is a violation of the NSHAP Data Use Agreement (DUA) to email NSHAP data in any form. *Do not do this!* If you have any questions about whether sending certain information is acceptable, please ask first. In general, statistical summaries or results (including plots) are fine, while any listing of individual data values (even for just a few respondents) is not. If you have a question about the data that can only be answered by discussing specific data values, please contact the list owner(s) at nshap-data-request@lists.uchicago.edu and a member of the NSHAP research team will contact you directly.

The NSHAP research team has established an email listserve for open questions and discussion about the dataset. Anyone who has obtained the dataset through the National Archive of Computerized Data on Aging (NACDA) is welcome to participate. To join the list, go to https://lists.uchicago.edu/web/subscribe/nshap-data and enter your email address. Someone from the research team will then contact you.

The scope of the list is broad, and any question about the dataset or about how to work with or analyze it is appropriate. However, please keep in mind that taking the time to think about your question and to compose it clearly and with sufficient detail will increase the chances that you will receive a helpful answer. That said, *do not under any circumstances send to the list information about data values for individual respondents*. If you have a question that requires referring to individual observations and/or data values, please contact the list owner(s) at nshap-data-request@lists.uchicago.edu and a member of the research team will contact you directly.

As the reader may know, there are several email listserves or discussion boards devoted to the discussion of specific software packages (e.g., Stata, R or SAS) and/or specific analytic techniques (e.g., Structural Equation Modeling). If your question is focused primarily on how to use a particular piece of software or how to perform a specific analysis, please do not be offended if you are referred to one of those lists (though it never hurts to ask!). However, we shall definitely try to answer any questions that are specifically relevant to the NSHAP dataset, since other NSHAP users are likely to benefit from the answer(s) as well.

CHAPTER

TWO

# MULTUM DRUG CATEGORY HIERARCHY

**Contents**

## 2.1 Introduction

This file describes Multum's hierarchy of therapeutic categories, and shows where items that are not included in the Multum database (i.e., those that have been mapped to a *Multum-extended drug name*) have been placed. See the Medications section of the `README` file for additional information.

## 2.2 Multum Therapeutic Classification

### 2.2.1 Alternative medicines

- herbal products

  **Note:** The following *Multum-extended drug names* have been placed in this category:

  - alfalfa
  - aloe vera
  - apple vinegar
  - astragalus
  - banisteria caapi
  - basil
  - borage
  - bromelain
  - burdock root
  - cayenne
  - cherry extract
  - cinnamon
  - cod liver oil
  - cordyceps
  - cranberry/blueberry/ascorbic acid
  - curcumin
  - fish-flax-borage
  - garlic-parsley
  - ginseng-muira puama-stevia
  - graviola
  - gymnema sylvestre
  - ivy extract
  - kelp
  - kelp, lecithin, pyridoxine
  - lutein-bilberry

- male sex enhancement supplements
- menopause relief supplements
- noni
- other herbals
- phytosterols
- phytozon
- pomegranate
- prostate health supplements
- pumpkin seed oil
- red wine
- royal jelly
- soy products
- wheat grass
- willow bark

* nutraceutical products

**Note:** The following *Multum-extended drug names* have been placed in this category:

- alpha lipoic acid
- alpha lipoic acid-ascorbic acid-calcium carbonate
- ambrotose
- celadrin-methylsulfonylmethane
- cholestin
- chondroitin/trisoamine/magnesium
- enzymes
- glucosamine-hyaluronic acid
- hyaluronic acid
- lactobacillus acidophilus-other herbals
- linoleic acid
- methylsulfonylmethane-ascorbic acid, other
- methylsulfonylmethane-glucosamine
- phenylalanine
- polyphenols
- taurine
- tyrosine
- uncategorized aminoacid combinations
- unspecified aminoacids

### 2.2.2 Anti-infectives

- amebicides
- aminoglycosides
- anthelmintics
- antifungals
  - azole antifungals
  - echinocandins
  - miscellaneous antifungals
  - polyenes
- antimalarial agents
  - antimalarial combinations
  - antimalarial quinolines
  - miscellaneous antimalarials
- antituberculosis agents
  - aminosalicylates
  - antituberculosis combinations
  - miscellaneous antituberculosis agents
  - nicotinic acid derivatives
  - rifamycin derivatives
  - streptomyces derivatives
- antiviral agents
  - NNRTIs
  - NRTIs
  - adamantane antivirals
  - antiviral combinations
  - antiviral interferons
  - miscellaneous antivirals
  - neuraminidase inhibitors
  - protease inhibitors
  - purine nucleosides
- carbapenems
- cephalosporins
  - first generation cephalosporins
  - fourth generation cephalosporins

    - second generation cephalosporins
    - third generation cephalosporins
- glycylcyclines
- leprostatics
- lincomycin derivatives
- macrolide derivatives
    - ketolides
    - macrolides
- miscellaneous antibiotics
- penicillins
    - aminopenicillins
    - antipseudomonal penicillins
    - beta-lactamase inhibitors
    - natural penicillins
    - penicillinase resistant penicillins
- quinolones
- sulfonamides
- tetracyclines
- urinary anti-infectives

### 2.2.3 Antihyperlipidemic agents

- HMG-CoA reductase inhibitors
- antihyperlipidemic combinations
- bile acid sequestrants
- cholesterol absorption inhibitors
- fibric acid derivatives
- miscellaneous antihyperlipidemic agents

### 2.2.4 Antineoplastics

- alkylating agents
- antibiotics/antineoplastics
- antimetabolites
- antineoplastic interferons
- antineoplastic monoclonal antibodies
- hormones/antineoplastics
- miscellaneous antineoplastics

- mitotic inhibitors

### 2.2.5 Biologicals

- antitoxins and antivenins
- colony stimulating factors
- in vivo diagnostic biologicals
- recombinant human erythropoietins

### 2.2.6 Cardiovascular agents

- agents for hypertensive emergencies
- agents for pulmonary hypertension
- angiotensin II inhibitors
- angiotensin converting enzyme inhibitors
- antiadrenergic agents, centrally acting
- antiadrenergic agents, peripherally acting
- antianginal agents
- antiarrhythmic agents
- antihypertensive combinations
- beta-adrenergic blocking agents
  - cardioselective beta blockers
  - non-cardioselective beta blockers
- calcium channel blocking agents
- diuretics
  - carbonic anhydrase inhibitors
  - loop diuretics
  - miscellaneous diuretics
  - potassium-sparing diuretics
  - thiazide diuretics
- inotropic agents
- miscellaneous cardiovascular agents
- peripheral vasodilators
- sclerosing agents
- vasodilators
- vasopressin antagonists
- vasopressors

### 2.2.7 Central nervous system agents

- analgesics
  - analgesic combinations
  - antimigraine agents
  - cox-2 inhibitors

    **Note:** The following *Multum-extended drug names* have been placed in this category:

    * etoricoxib (experimental)

  - miscellaneous analgesics
  - narcotic analgesic combinations
  - narcotic analgesics
  - nonsteroidal anti-inflammatory agents
  - salicylates
- anorexiants
- anticonvulsants
  - barbiturate anticonvulsants
  - benzodiazepine anticonvulsants

    **Note:** The following *Multum-extended drug names* have been placed in this category:

    * frisium (foreign form benzodiazepine)

  - dibenzazepine anticonvulsants
  - hydantoin anticonvulsants
  - miscellaneous anticonvulsants
  - oxazolidinedione anticonvulsants
  - succinimide anticonvulsants
- antiemetic/antivertigo agents
  - 5HT3 receptor antagonists
  - anticholinergic antiemetics
  - miscellaneous antiemetics
  - phenothiazine antiemetics
- antiparkinson agents
  - anticholinergic antiparkinson agents
  - dopaminergic antiparkinsonism agents
  - miscellaneous antiparkinson agents
- cholinergic agonists

- cholinesterase inhibitors
- general anesthetics
- miscellaneous central nervous system agents
- muscle relaxants
  - neuromuscular blocking agents
  - skeletal muscle relaxant combinations
  - skeletal muscle relaxants

### 2.2.8 Coagulation modifiers

- anticoagulants
  - coumarins and indandiones
  - factor Xa inhibitors
  - heparins
  - thrombin inhibitors
- antiplatelet agents
  - glycoprotein platelet inhibitors
  - platelet aggregation inhibitors
- heparin antagonists
- miscellaneous coagulation modifiers
- thrombolytics

### 2.2.9 Gastrointestinal agents

- 5-aminosalicylates
- GI stimulants
- H2 antagonists
- antacids
- anticholinergics/antispasmodics
- antidiarrheals
- digestive enzymes
- gallstone solubilizing agents
- laxatives
- miscellaneous GI agents
- proton pump inhibitors

### 2.2.10 Hormones

- adrenal cortical steroids
  - corticotropin
  - glucocorticoids
  - mineralocorticoids
- antidiabetic agents
  - alpha-glucosidase inhibitors
  - antidiabetic combinations
  - insulin
  - meglitinides
  - miscellaneous antidiabetic agents
  - non-sulfonylureas
  - sulfonylureas
  - thiazolidinediones
- bisphosphonates
- growth hormones
- miscellaneous hormones
- sex hormones
  - 5-alpha-reductase inhibitors
  - androgens and anabolic steroids
  - contraceptives
  - estrogens
  - gonadotropin releasing hormones
  - gonadotropins
  - miscellaneous sex hormones
  - progestins
  - sex hormone combinations
- thyroid drugs

**Note:** The following *Multum-extended drug names* have been placed in this category:

- cynoplus (foreign form thyroid desiccated and levothyroxine)

### 2.2.11 Immunologic agents

- bacterial vaccines
- immune globulins
- immunosuppressive agents
- interferons
- miscellaneous biologicals
- monoclonal antibodies
- toxoids
- viral vaccines

### 2.2.12 Miscellaneous agents

- antidotes
- antigout agents
- antihyperuricemic agents
- antipsoriatics
- antirheumatics
- chelating agents
- cholinergic muscle stimulants
- genitourinary tract agents
  - impotence agents
  - miscellaneous genitourinary tract agents
  - urinary antispasmodics
  - urinary pH modifiers
- glucose elevating agents
- illicit (street) drugs
- insulin-like growth factor
- local injectable anesthetics
- miscellaneous uncategorized agents
- psoralens
- smoking cessation agents
- viscosupplementation agents

### 2.2.13 Nutritional products

- intravenous nutritional products
- iron products

**Note:** The following *Multum-extended drug names* have been placed in this category:

  - ascorbic acid/ferrous fumarate/cyanocobalamin/folic acid
  - ascorbic acid/ferrous fumarate/cyanocobalamin
  - unspecified iron

- minerals and electrolytes

**Note:** The following *Multum-extended drug names* have been placed in this category:

  - calcium unspecified
  - calcium/magnesium/zinc
  - unspecified magnesium
  - unspecified manganese
  - unspecified potassium
  - zinc

- oral nutritional supplements
- vitamin and mineral combinations

**Note:** The following *Multum-extended drug names* have been placed in this category:

  - antioxidants
  - ascorbic acid-calcium carbonate
  - ascorbic acid-rosehips
  - ascorbic acid-vitamin a-vitamin e
  - ascorbic acid-vitamin e
  - ascorbic acid-vitamin e-calcium
  - ascorbic acid/ferrous fumarate/cyanocobalamin
  - ascorbic acid/pyridoxine/zinc
  - ascorbic acid/selenium/pyridoxine/zinc
  - calcium/magnesium/vitamin d
  - calcium/magnesium/vitamin d/boron
  - calcium/vitamin d/zinc
  - cyanocobalamin/folic acid/pyridoxine
  - eye vitamins

  - lutein-zeaxanthin
  - multivitamin with minerals-ginseng
  - multivitamin, other
  - multivitamin, stress
  - multivitamin,lutein
  - niacinamide-folic acid-zinc
  - pyridoxine/magnesium/potassium
  - supplement for cardiovacular health
  - thiamin-niacin
  - uncategorized vitamin/herbal combinations
  - vitamin a-vitamin d
  - vitamin b-complex
  - vitamin b-complex and ascorbic acid
  - vitamin b-complex and folic acid
  - women's bone health

- vitamins

  **Note:** The following *Multum-extended drug names* have been placed in this category:

  - biotin
  - lutein
  - unspecified vitamin b

### 2.2.14 Other supplements

This category does not appear in the official Multum hierarchy. It was added to accommodate those vitamins, herbal and alternative medicines reported by NSHAP respondents that are not included in the Multum database.

The following *Multum-extended drug names* have been placed in this category:

- fiber
- uncategorized supplements
- weightloss

### 2.2.15 Plasma expanders

### 2.2.16 Psychotherapeutic agents

- CNS stimulants
- antidepressants
  - SSNRI antidepressants
  - SSRI antidepressants
  - miscellaneous antidepressants
  - monoamine oxidase inhibitors
  - phenylpiperazine antidepressants
  - tetracyclic antidepressants
  - tricyclic antidepressants
- antipsychotics
  - miscellaneous antipsychotic agents
  - phenothiazine antipsychotics
  - psychotherapeutic combinations
  - thioxanthenes
- anxiolytics, sedatives, and hypnotics
  - barbiturates
  - benzodiazepines

    **Note:** The following *Multum-extended drug names* have been placed in this category:

    * frisium (foreign form benzodiazepine)

  - miscellaneous anxiolytics, sedatives and hypnotics

### 2.2.17 Radiologic agents

- radiocontrast agents
  - iodinated contrast media
  - lymphatic staining agents
  - magnetic resonance imaging contrast media
  - non-iodinated contrast media
  - ultrasound contrast media
- radiologic adjuncts
- radiopharmaceuticals
  - diagnostic radiopharmaceuticals
  - therapeutic radiopharmaceuticals

### 2.2.18 Respiratory agents

- antiasthmatic combinations
- antihistamines
- antitussives
- bronchodilators
  - adrenergic bronchodilators
  - anticholinergic bronchodilators
  - bronchodilator combinations
  - methylxanthines
- decongestants
- expectorants
- leukotriene modifiers
- lung surfactants
- miscellaneous respiratory agents
- respiratory inhalant products
  - inhaled corticosteroids
  - mast cell stabilizers
  - mucolytics
- upper respiratory combinations

### 2.2.19 Topical agents

- anorectal preparations
- antiseptic and germicides
- dermatological agents
  - miscellaneous topical agents
  - topical acne agents
  - topical anesthetics
  - topical anti-infectives
  - topical antibiotics
  - topical antifungals
  - topical antipsoriatics
  - topical antivirals
  - topical emollients
  - topical steroids
  - topical steroids with anti-infectives

- mouth and throat products
- nasal preparations
  - nasal antihistamines and decongestants
  - nasal lubricants and irrigations
  - nasal steroids
- ophthalmic preparations
  - miscellaneous ophthalmic agents
  - mydriatics
  - ophthalmic anesthetics
  - ophthalmic anti-infectives
  - ophthalmic anti-inflammatory agents
  - ophthalmic antihistamines and decongestants
  - ophthalmic diagnostic agents
  - ophthalmic glaucoma agents
  - ophthalmic lubricants and irrigations
  - ophthalmic steroids
  - ophthalmic steroids with anti-infectives
  - ophthalmic surgical agents
- otic preparations
  - miscellaneous otic agents
  - otic anti-infectives
  - otic steroids with anti-infectives
- sterile irrigating solutions
- vaginal preparations
  - miscellaneous vaginal agents
  - spermicides
  - vaginal anti-infectives

CHAPTER

THREE

# HISTORY OF CHANGES

## Version 3.0

New data added:

1) Data from COVID-19 substudy performed in Fall 2020
2) Resilience items added in Wave 3 (`bounce`, `energetic`, `stride` and `setmind`); these were asked in the leave-behind questionnaire
3) Items regarding Internet use were added for Wave 2 (`accessweb`, `webhome` and `webuse`) and Wave 3 (`iemail`)
4) Added year of birth for Waves 1-3 (`ageyear`)

Other changes:

1) Corrected year of birth for 10027680; note that this respondent was 56 at the time of the Wave 1 interview.
2) Updates to network data for all three waves based on careful review of matching across waves

## Version 3.0b3

New data added:

1) Wave 3 weights adjusted for non-response
2) Wave 3 design variables (`stratum` and `cluster`) to permit calculation of design-based variance estimates
3) Wave 3 proxy interviews

Other changes:

1) Selection weights (`weight_sel`) have been updated
2) Added Wave 3 to dataset available via direct download

## Version 3.0b2

New data added:

1) Wave 3 selection weights (without adjustment for non-response)
2) Wave 3 modules: Sexual Activity, Sexual Interest, Interviewer Comments
3) Wave 2 couple weights

Other changes:

1) First release of dataset available via direct download from NACDA (Waves 1 and 2 only); see section on *Direct Download Dataset* in data documentation.

## Version 3.0b1

New data added:

1) Initial data from Wave 3!
2) Additional data from Wave 2, including medications, proxy interview and actigraphy sleep variables (nightly summaries)
3) Added vars `lastprd` and `agelstpd` (Wave 2)
4) Vaginal cell cytology (Wave 1)

Other changes:

1) Starting in Wave 2, variables `ccfm_*` from the MoCA-SA cognitive function measure have been renamed `moca_*`
2) Starting in Wave 2, variable `atndserv` was renamed to `atndserv2` to acknowledge a change in the response categories from Wave 1. In addition, the variable was recoded from:

```
0 never
2 about once or twice a year
3 several times a year
4 about once a month
5 every week
6 several times a week
```

to:

```
0 never
1 about once or twice a year
2 several times a year
3 about once a month
4 every week
5 several times a week
```

to close the gap caused by the response option that was removed.

# Version 2.2

**Note:** The Wave 2 person level weights (variables weight_sel and weight_adj in the W2 core file) have been updated. These differ from the original Wave 2 weights in how they treat the Wave 1 nonrespondents, and will yield more precise estimates (i.e., smaller standard errors). They should be used in preference to the Wave 2 weights in previous data releases.

Since a corresponding Wave 2 weight for W1 age-eligibles only is not yet available, the previous variable weight_ageelig has been (temporarily) dropped. Analysts wishing to restrict their analyses to W1 age-eligibles may do so using the variable ageelig.

New data added:

1) For Wave 2, added SAQ data for respondents who answered on paper.
2) Data from Interviewer Comments module for Wave 2 have been added.
3) Data from Wave 2 questions on sleep meds have been added.
4) Dried blood spot assays have been added, including CRP, Epstein-Barr virus, total hemoglobin and glycosylated hemoglobin (A1c).
5) Saliva (passive drool) steroid assays have been added, including testosterone, DHEA, estradiol and progesterone.
6) Data from Wave 2 questions on sexual interest, sexual activity and pre-pubertal sexual experiences have been added (including items from leave-behind).
7) Remaining data from leave-behind including items on childhood background, social activity, bereavement, neighborhood, caregiving and physical contact have been added.

Other changes:

1) Backcoded response to sml_intro2 from "continue" to "refused" for 8 respondents who declined to particate in any part of the olfaction module (as determined by a review of the interviewer notes). Also added variables redpen2_flag, greenpen2_flag and bluepen_flag denoting additional respondents who declined one or more of the three olfaction tasks (again based on a review of the interviewer notes).
2) For 27 respondents in Wave 1, changed values for SAQ items mensex, womensex, paysex, mstbate, mstbateo, urinepr, othurine and stoolinc from "missing in error" to "not returned". These respondents agreed to answer the SAQ on paper, but either did not return it to the interviewer or it was not returned to NORC.
3) Wave 2 vars famopen2 and fropen2 were recoded so that the responses "if volunteered - no family" and "if volunteered - no friends", previously encoded as -1, are now encoded as 4 (all other responses remain encoded as before). This change was made to avoid confusion for those who may wish to recode extended missing values as negative integers.

# Version 2.1

New data added:

1) The remainder of the questionnaire items from Wave 2, including the network roster, spouse/partner history, and leave-behind. This also includes the cognitive function module and those biomeasures recorded during the interview (e.g., anthropometrics, blood pressure, timed walk and chair stands).

Other changes:

1) Changed `disp_grp` (Wave 2) from "poor health" to "other - known/presumed alive" for 4 respondents, based on re-review of final disposition codes.
2) Renamed the following variables for consistency with standard SPMSQ administration/scoring sheet:
   - `spmsq_ans5` renamed to `spmsq_ans4a`
   - `spmsq_ans6` renamed to `spmsq_ans5`
   - `spmsq_ans7a` renamed to `spmsq_ans6a`
   - `spmsq_ans7b` renamed to `spmsq_ans6b`
   - `spmsq_ans7c` renamed to `spmsq_ans6c`
   - `spmsq_ans7` renamed to `spmsq_ans6`
   - `spmsq_ans8` renamed to `spmsq_ans7`
   - `spmsq_ans9` renamed to `spmsq_ans8`
   - `spmsq_ans10` renamed to `spmsq_ans9`
   - `spmsq_ans11` renamed to `spmsq_ans10`

## Version 2.0

New data added:

1) Wave 2 data from interview and leave-behind questionnaire
2) New variable for Wave 1:

   **int_start** Month and year in which in-person interview was started (W1 and W2)

Other changes:

1) Based on the W2 data, it was determined that 10041240 had been erroneously classified as male during W1; the gender for this respondent has been changed to female.
2) The stratum and cluster assignments for two respondents (10008510 and 10034960) were incorrect (their values had been flipped). This has been fixed in both the W1 and W2 datasets.

## Version 1.6

New data added:

1) Political affiliation (leave-behind)
2) Elder mistreatment module (CAPI and leave-behind)
3) Cognitive function module (SPMSQ)

Other changes:

1) The unit of measurement for hemoglobin (var hb) was incorrectly specified as mg/dL. This has been changed to g/dL.

# Version 1.5

New data added:

1) Bereavement items (leave-behind)
2) Social activity items (leave-behind)
3) Physical contact module (CAPI and leave-behind)
4) Sexual interest module (CAPI and leave-behind)
5) Get Up and Go assessment of physical function
6) Results for bacterial vaginosis and candidiasis from vaginal swabs

Other changes:

7) A thorough review of the marital and cohabitational history data was undertaken in order to identify duplicates. The following 26 duplicate records were found and deleted from the file `nshap-maritalhistory`:

   - 10003650 (scid 202)
   - 10004780 (scid 201)
   - 10005010 (scid 201)
   - 10014750 (scid 202)
   - 10016900 (scid 203)
   - 10018600 (scid 202)
   - 10018610 (scid 202)
   - 10010990 (scid 201)
   - 10020300 (scid 201)
   - 10020780 (scid 201)
   - 10024090 (scid 201)
   - 10028570 (scid 206)
   - 10030420 (scid 201)
   - 10030770 (scid 202)
   - 10031190 (scid 201 and 202)
   - 10034250 (scid 203, 204, 205 and 206)
   - 10035510 (scid 201)
   - 10036550 (scid 202)
   - 10037000 (scid 201)
   - 10040570 (scid 201)
   - 10042520 (scid 201)
   - 10041800 (scid 201)

   Note that this also affects the variables `livec` and `numlivec` in the core file. See the section on Marital/Cohabiting History in the `README.txt` file for more information.

8) Added technical report on blood spot assays

# Version 1.4

New data added:

1) Nutritional products and alternative medicines identified in the medication log have now been matched to their corresponding Multum drug name, or, if not covered by Multum, have been mapped to a set of drug names designed to extend the Multum database. Entries which could not be mapped to a specific drug name have either been coded according to their therapeutic value (e.g., "medication for my blood pressure") or are identified as uncodeable.
2) Added data on sexual partner history during the past 5 years (Boxes A and B of Section 3A)
3) The month and year on which the interview took place has been added, to permit users to work with the dates in the marital, cohab, and sexual partner histories relative to the date of the interview
4) Added data from biomeasures of smell, taste, distance vision, and touch
5) Added results from assays of blood spots, including C-reactive protein, Epstein-Barr virus, hemoglobin, and glycosylated hemoglobin (HbA1c)

Other changes:

6) In performing (1) above, a few minor changes were made to the existing matching of individual entries to Multum drug names. As a result, some of the existing medication-related variables have changed for a small number of cases. Note that the Medications section in the `README` file has been updated to reflect these changes.
7) The file `multum_hierarchy.txt` has been updated to show how the extended drug names introduced in (1) have been classified.
8) Added technical reports describing the salivary assays and measurement of bacterial vaginosis, HPV, and vaginal candidiasis from the vaginal swabs

# Version 1.3

New data added:

1) Added data on religion (Section 8)

Other changes:

2) The .dta files in version 1.2 were inadvertently saved as Stata 10 datasets. Since some users may not yet have upgraded to Stata 10, and since we do not currently use any features which require the new format, we have gone back to saving .dta files in Stata 9 format.

# Version 1.2

New data added:

1) Added data recorded by interviewer following interview (including ratings of respondent with respect to several characteristics)
2) Added data from caregiving items asked in leave-behind
3) Added data on employment and finances (Section 7)

Other changes:

4) Variable cluster previously used the values 1 and 2 to distinguish between the two pseudo-PSUs within each stratum; these have been replaced with unique cluster identifiers to prevent inadvertent collapsing of clusters when collapsing strata

5) The core file has been setup using the appropriate svyset command

# Version 1.1

New data added:

1) Added data from questions on sexual attitudes (asked in leave-behind only)

2) Added data from saliva assays including cotinine, DHEA, estradiol, progesterone, and testosterone

3) Added data from the medication log, including therapeutic indicators derived by linking individual medications to the Multum database

Other changes:

4) Dropped confirm variable from network file, which recorded technical details of the interview and was useful only for data cleaning

5) Values for height and waistm appear to have been reversed for 10003030 and 10030780 and were originally set to missing since they appeared as gross outliers in the joint distribution of height, weight, and waistm; these values have now been restored into the appropriate positions

6) Recovered race for one additional respondent by backcoding based on open-ended response. Note that the derived variable *ethgrp* has been updated to reflect this change.

7) Added new variable called degree_coded; this is identical to the original variable degree in all cases where a specific category was used, but in cases where degree equals “other” has been backcoded based on the corresponding open ended responses and the other education variables. Note that the derived variable educ has been updated to reflect this change.

8) Changed value of hschl from “no” to “yes, equivalency” for two respondents based on their responses to degree_oth

9) Removed variables indicating self-reported HIV status (conditns_10, havestds_9, and flareups_9)

10) Restored variables indicating month of specific events, including beginning and ending of marriage or cohabitation, cancer diagnosis, and last time sex partner had an affair)

# Version 1.0

Initial version submitted to NACDA.

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